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Semiconductor package

Abstract

A semiconductor package includes a first substrate, a second substrate provided on the first substrate, a semiconductor chip provided between the first substrate and the second substrate, solder structures extending between the first substrate and the second substrate and spaced apart from the semiconductor chip, and bumps provided between the semiconductor chip and the second substrate. The solder structures electrically connect the first substrate and the second substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/032,916 filed on Sep. 25, 2020, which is a continuation of U.S. application Ser. No. 16/424,000 filed on May 28, 2019, now U.S. Pat. No. 10,825,774, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2018-0089681, filed on Aug. 1, 2018, and Korean Patent Application No. 10-2018-0139720, filed on Nov. 14, 2018, in the Korean Intellectual Property Office, the entire contents of which are hereby incorporated by reference.

BACKGROUND

(1) The present disclosure relates to a semiconductor package, and in particular, to a stack-type semiconductor package.

(2) In the case where an integrated circuit (IC) chip is provided in the form of a semiconductor package, it can be easily used as a part of an electronic product. In general, the semiconductor package includes a printed circuit board (PCB) and a semiconductor chip, which is mounted on the

PCB and is electrically connected to the PCB by bonding wires or bumps. With development of the semiconductor industry, many studies are being conducted to improve reliability of the semiconductor package.

SUMMARY

(3) One or more example embodiments provide a semiconductor package having a minimized number of defects and improved reliability and a method of fabricating the same.

(4) According to an aspect of an example embodiment, a semiconductor package may include a first substrate, a second substrate provided on the first substrate, a semiconductor chip provided between the first substrate and the second substrate, solder structures extending between the first substrate and the second substrate and spaced apart from the semiconductor chip, the solder structures electrically connecting the first substrate and the second substrate, and bumps provided between the semiconductor chip and the second substrate.

(5) According to an aspect of an example embodiment, a semiconductor package may include a first substrate, a second substrate provided on the first substrate, a first semiconductor chip provided between the first substrate and the second substrate, solder structures provided between the first substrate and the second substrate and spaced apart from the first semiconductor chip, the solder structures electrically connecting the first substrate and the second substrate, and supporting patterns provided between the first substrate and the second substrate and spaced apart from the first semiconductor chip, wherein the solder structures are disposed between the first semiconductor chip and the supporting patterns, and the supporting patterns enclose sides of the first semiconductor chip.

(6) According to an aspect of an example embodiment, a semiconductor package may include a first substrate, a second substrate provided on the first substrate, a semiconductor chip provided between the first substrate and the second substrate, bumps provided between the semiconductor chip and the second substrate and supporting the second substrate to be spaced apart from the semiconductor chip, and solder structures extending between the first substrate and the second substrate and spaced apart from the semiconductor chip, wherein each of the solder structures includes a lower solder ball provided on an upper surface of the first substrate, and an upper solder ball provided on a lower surface of the second substrate and reflow joined to the lower solder ball

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Example embodiments will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:

(2) FIG. 1 is a plan view illustrating a first semiconductor package according to an example embodiment, and FIG. 2 is a sectional view taken along a line I-I' of FIG. 1;

(3) FIGS. 3A to 3D are plan views illustrating some modified structures of a guide pattern of FIG. 1;

(4) FIGS. 4 and 5 are sectional views illustrating a method of fabricating a first semiconductor package according to an example embodiment, taken along a line corresponding to the line I-I' of FIG. 1;

(5) FIG. 6 is an enlarged view of a portion 'A' of FIG. 4;

(6) FIGS. 7 and 8 are sectional views illustrating a method of fabricating a first semiconductor package according to an example embodiment, taken along a line corresponding to the line I-I' of FIG. 1;

(7) FIG. 9 is a sectional view illustrating a stack-type semiconductor package according to an example embodiment;

(8) FIG. 10 is a sectional view illustrating a first semiconductor package according to an example

embodiment, taken along a line corresponding to the line I-I' of FIG. 1.

(9) FIG. 11 is a sectional view illustrating a first semiconductor package according to an example embodiment, taken along a line corresponding to the line I-I' of FIG. 1;

(10) FIG. 12 is a plan view illustrating a first semiconductor package according to an example embodiment, and FIG. 13 is a sectional view taken along a line I-I' of FIG. 12;

(11) FIG. 14 is a sectional view illustrating a stack-type semiconductor package according to an example embodiment.

(12) FIG. 15 is a plan view illustrating a first semiconductor package according to an example embodiment, and FIG. 16 is a sectional view taken along a line I-I' of FIG. 15;

(13) FIG. 17 is an enlarged view of a portion 'AA' of FIG. 16;

(14) FIG. 18 is a sectional view illustrating a modified structure of a first semiconductor package according to an example embodiment, taken along a line I-I' of FIG. 15;

(15) FIG. 19 is an enlarged view of a portion a portion 'BB' of FIG. 18;

(16) FIG. 20 is a plan view illustrating a method of fabricating a first semiconductor package according to an example embodiment;

(17) FIGS. 21 to 24 are sectional views illustrating a method of fabricating a first semiconductor package according to an example embodiment, taken along a line corresponding to a line I-I' of FIG. 20;

(18) FIG. 25 is an enlarged view of a portion CC of FIG. 21, FIG. 26 is an enlarged view of a portion DD of FIG. 23, and FIG. 27 is an enlarged view of a portion EE of FIG. 24;

(19) FIG. 28 is a sectional view illustrating a method of fabricating a first semiconductor package according to a modified example embodiment, taken along a line corresponding to the line I-I' of FIG. 20;

(20) FIG. 29 is a plan view illustrating a method of fabricating a first semiconductor package according to an example embodiment;

(21) FIGS. 30 to 33 are sectional views illustrating a method of fabricating a first semiconductor package according to an example embodiment, taken along a line corresponding to a line I-I' of FIG. 29; and

(22) FIG. 34 is a sectional view illustrating a stack-type semiconductor package according to an example embodiment.

DETAILED DESCRIPTION

(23) Example embodiments will now be described more fully with reference to the accompanying drawings.

(24) It should be noted that the drawings are intended to illustrate the general characteristics of methods, structure and/or materials utilized in certain example embodiments and to supplement the written description provided below. These drawings are not, however, to scale and may not precisely reflect the precise structural or performance characteristics of any given embodiment, and should not be interpreted as defining or limiting the range of values or properties encompassed by example embodiments. For example, the relative thicknesses and positioning of layers, regions and/or structural elements may be reduced or exaggerated for clarity. The use of similar or identical reference numbers in the various drawings is intended to indicate the presence of a similar or identical element or feature.

(25) FIG. 1 is a plan view illustrating a first semiconductor package P1 according to an example embodiment. FIG. 2 is a sectional view taken along a line I-I' of FIG. 1.

(26) Referring to FIGS. 1 and 2, a first semiconductor chip 200 and a first mold layer 300 may be provided on a first substrate 100. The first substrate 100 may be a printed circuit board with a circuit pattern or a redistribution layer. The first substrate 100 may include first substrate pads 110 and second substrate pads 120. The first substrate pads 110 and the second substrate pads 120 may be respectively provided on an upper surface 100U and a lower surface 100L of the first substrate 100. The second substrate pads 120 may be connected to corresponding pads of the first substrate

pads **110** through internal interconnection lines. In the following diagrams, the internal interconnection line in the first substrate **100** will be schematically illustrated by a dotted line. The first substrate pads **110** and the second substrate pads **120** may be formed of or include at least one of conductive materials. Outer terminals **130** may be provided on the lower surface **100L** of the first substrate **100** and may be connected to the second substrate pads **120**. The outer terminal **130** may be formed of or include at least one of conductive material and may be provided in the form of a solder ball.

(27) The first semiconductor chip **200** may include chip pads **210** provided on a lower surface thereof. Connecting portions **220** may be provided between the first substrate **100** and the first semiconductor chip **200** and may be connected to the chip pads **210**, respectively. The connecting portions **220** may be connected to corresponding pads of the first substrate pads **110**. The first semiconductor chip **200** may be electrically connected to the first substrate **100** through the connecting portions **220**. The chip pads **210** and the connecting portions **220** may be formed of or include at least one of conductive materials. Each of the connecting portions **220** may include one of solder balls, bumps, or pillars. The first semiconductor chip **200** may include integrated circuits and, in an example embodiment, the integrated circuits may be adjacent to the lower surface of the first semiconductor chip **200**. The first semiconductor chip **200** may be a logic chip. As an example, the first semiconductor chip **200** may be a non-memory chip (e.g., an application processor).

(28) The first mold layer **300** may cover the upper surface **100U** of the first substrate **100** and a side surface **200S** of the first semiconductor chip **200**. The first mold layer **300** may be provided to fill a gap region between the first substrate **100** and the first semiconductor chip **200** and to seal the connecting portions **220**. The first mold layer **300** may be provided to expose an upper surface **200U** of the first semiconductor chip **200**. An upper surface **300U** of the first mold layer **300** may be located at substantially the same height as the upper surface **200U** of the first semiconductor chip **200**. In the present specification, the term “height” will be used to refer to a distance measured from the upper surface **100U** of the first substrate **100**. The first mold layer **300** may include at least one of epoxy molding compounds.

(29) The first mold layer **300** may include openings **310**. Each of the openings **310** may penetrate through the first mold layer **300** and expose a corresponding one of the first substrate pads **110**. The openings **310** may be horizontally spaced apart from the first semiconductor chip **200**. In other words, the openings **310** may be spaced apart from the first semiconductor chip **200** in a direction parallel to the upper surface **100U** of the first substrate **100**. Solder structures **500** may be provided in the openings **310**, respectively. The solder structures **500** may be connected to corresponding pads of the first substrate pads **110**. Each of the solder structures **500** may be electrically connected to the outer terminal **130** or the first semiconductor chip **200**. The solder structures **500** may be formed of or include at least one of conductive materials (e.g., tin, lead, silver, or alloys thereof).

(30) A plurality of bumps **420** may be provided on the upper surface **200U** of the first semiconductor chip **200**. The bumps **420** may be horizontally spaced apart from each other. For example, the bumps **420** may be spaced apart from each other in a direction parallel to the upper surface **100U** of the first substrate **100**. The bumps **420** may be two-dimensionally arranged in two different directions parallel to the upper surface **100U** of the first substrate **100**, but embodiments are not limited to this example. Each of the bumps **420** may have a pillar shape and protrude from the upper surface **200U** of the first semiconductor chip **200**. Each of the bumps **420** may have a length **420H**, when measured in a direction perpendicular to the upper surface **100U** of the first substrate **100**. In some example embodiments, the bumps **420** may be formed of or include a material different from the first mold layer **300**. As an example, the bumps **420** may include at least one of insulating polymer materials, conductive polymer materials, or solder resist materials. In certain example embodiments, the bumps **420** may be formed of or include the same material as the first mold layer **300**.

(31) A guide pattern **410** may be provided on the first substrate **100** and may be horizontally spaced apart from the first semiconductor chip **200**. For example, the guide pattern **410** may be spaced apart from the first semiconductor chip **200** in a direction parallel to the upper surface **100U** of the first substrate **100**. In some example embodiments, the guide pattern **410** may be provided on the upper surface **300U** of the first mold layer **300**. When viewed in a plan view, the guide pattern **410** may be provided to enclose the side surface **200S** of the first semiconductor chip **200** and the solder structures **500** may be provided between the first semiconductor chip **200** and the guide pattern **410**. As an example, the guide pattern **410** may be extended along the side surface **200S** of the first semiconductor chip **200** and may have a polygonal shape, when viewed in a plan view. Some modified examples of the planar shape of the guide pattern **410** will be described in more detail with reference to FIGS. 3A to 3D.

(32) The guide pattern **410** may protrude from the upper surface **300U** of the first mold layer **300**. The guide pattern **410** may have a length **410H** in the direction perpendicular to the upper surface **100U** of the first substrate **100**. The length **410H** of the guide pattern **410** may be greater than the length **420H** of each of the bumps **420**. An upper surface **410U** of the guide pattern **410** may be positioned at a height higher than a height of the upper surfaces **420U** of the bumps **420**. In some example embodiments, the guide pattern **410** may be formed of or include a material different from the first mold layer **300**. As an example, the guide pattern **410** may include at least one of insulating polymer materials, conductive polymer materials, or solder resist materials. In certain example embodiments, the guide pattern **410** may be formed of or include the same material as the first mold layer **300**. The guide pattern **410** may be formed of or include the same material as the bumps **420**.

(33) A second substrate **600** may be provided on the first semiconductor chip **200** and the first mold layer **300**. In some example embodiments, the second substrate **600** may include an interposer substrate. In this case, the second substrate **600** may be formed of or include at least one of insulating resins. As an example, the insulating resin may include a solder resist material (e.g., photosensitive polyimide), but embodiments are not limited to this example. First pads **610** and second pads **620** may be respectively provided on an upper surface **600U** and a lower surface **600L** of the second substrate **600**. Interconnection lines may be provided in the second substrate **600** to electrically connect the first pads **610** and the second pads **620** to each other. In FIG. 2, the interconnection lines are schematically illustrated by the dotted line in the second substrate **600**. The first pads **610** and the second pads **620** may be formed of or include at least one of conductive or metallic materials (e.g., copper or aluminum). The solder structures **500** may be connected to the second pads **620**. The second substrate **600** may be electrically connected to the first substrate **100** and the first semiconductor chip **200** through the solder structures **500**.

(34) The bumps **420** may be interposed between the first semiconductor chip **200** and the second substrate **600**. The bumps **420** may be in contact with the upper surface **200U** of the first semiconductor chip **200** and the lower surface **600L** of the second substrate **600**. The guide pattern **410** may have inner side surfaces **410IS**, which are provided to face each other with the bumps **420** interposed therebetween. The second substrate **600** may be provided between the inner side surfaces **410IS** of the guide pattern **410**. When viewed in a plan view, the guide pattern **410** may be provided to enclose a side surface of the second substrate **600**. The upper surface **410U** of the guide pattern **410** may be located at a height higher than the lower surface **600L** of the second substrate **600**.

(35) An under-fill layer **550** may be interposed between the second substrate **600** and the first semiconductor chip **200** and between the second substrate **600** and the first mold layer **300**. The under-fill layer **550** may be extended into the openings **310** in the first mold layer **300** to cover the solder structures **500**. The under-fill layer **550** may be interposed between the bumps **420** and may be locally provided between the inner side surfaces **410IS** of the guide pattern **410**. The under-fill layer **550** may be formed of or include a material different from the first mold layer **300**. As an

example, the under-fill layer **550** may be formed of or include an insulating resin. In some example embodiments, the under-fill layer **550** may be omitted.

(36) FIGS. **3A** to **3D** are plan views illustrating some modified structures of the guide pattern **410** of FIG. **1**. To avoid complexity of the drawings, the solder structures **500** and the openings **310** are not illustrated in FIGS. **3A** to **3D**.

(37) Referring to FIGS. **2**, **3A**, and **3B**, the guide pattern **410** may include a plurality of segments **412** spaced apart from each other. The segments **412** may be provided on the upper surface **300U** of the first mold layer **300** and may be horizontally spaced apart from each other. The segments **412** may be provided to enclose the side surface **200S** of the first semiconductor chip **200**, when viewed in a plan view. The segments **412** may be arranged to be spaced apart from each other along the side surface **200S** of the first semiconductor chip **200**. The segments **412** may have inner side surfaces **412IS** facing each other with the bumps **420** interposed therebetween. The inner side surfaces **412IS** of the segments **412** may correspond to the inner side surfaces **410IS** of the guide pattern **410** shown in FIG. **2**. The second substrate **600** may be disposed between the inner side surfaces **412IS** of the segments **412**. When viewed in a plan view, the segments **412** may be provided to enclose a side surface of the second substrate **600**. As an example, as shown in FIG. **3A**, the segments **412** may be provided to cover portions of the side surface of the second substrate **600** and corners of the second substrate **600**. As another example, as shown in FIG. **3B**, the segments **412** may be provided to cover only the corners of the second substrate **600**.

(38) Referring to FIGS. **2** and **3C**, a plurality of holes **430** may be provided in the guide pattern **410**. Each of the holes **430** may penetrate at least a portion of the guide pattern **410**. The holes **430** may be spaced apart from each other along the side surface **200S** of the first semiconductor chip **200**, when viewed in a plan view. The holes **430** may be arranged to enclose the side surface of the second substrate **600**, when viewed in a plan view.

(39) Referring to FIGS. **2** and **3D**, the guide pattern **410** may include the segments **412**, and the segments **412** may be provided to cover a portion of the side surface of the second substrate **600** and corners of the second substrate **600**. At least one hole **430** may be provided in each of the segments **412**. The at least one hole **430** may penetrate at least a portion of each of the segments **412**. Except for the afore-described differences, the guide patterns **410** according to the modified example embodiments may be substantially the same as the guide pattern **410** described with reference to FIGS. **1** and **2**.

(40) FIGS. **4** and **5** are sectional views illustrating a method of fabricating the first semiconductor package **P1** according to an example embodiment, taken along a line corresponding to the line I-I' of FIG. **1**. FIG. **6** is an enlarged view of a portion 'A' of FIG. **4**. In order to avoid redundancy, overlapping description of the first semiconductor package **P1** described with reference to FIGS. **1**, **2** and **3A** to **3D** may be omitted.

(41) Referring to FIGS. **1** and **4**, the first semiconductor chip **200** may be mounted on the first substrate **100**. The first mold layer **300** may be formed on the first substrate **100** to cover the side surface **200S** of the first semiconductor chip **200**. The first mold layer **300** may be provided to expose an upper surface **200U** of the first semiconductor chip **200**. The openings **310** may be formed in the first mold layer **300**. Each of the openings **310** may penetrate through the first mold layer **300** and expose a corresponding one of the first substrate pads **110**. The formation of the openings **310** may include removing a portion of the first mold layer **300** using, for example, a laser drilling process. The openings **310** may be horizontally spaced apart from the first semiconductor chip **200**. Lower solder balls **510** may be formed in the openings **310**, respectively. The lower solder balls **510** may be electrically connected to the outer terminal **130** or the first semiconductor chip **200**. The lower solder balls **510** may be formed of or include at least one of conductive materials (e.g., tin, lead, silver, or alloys thereof).

(42) The bumps **420** may be formed on the upper surface **200U** of the first semiconductor chip **200**, and the guide pattern **410** may be formed on the upper surface **300U** of the first mold layer **300**.

When viewed in a plan view, the guide pattern **410** may enclose the side surface **200S** of the first semiconductor chip **200**. As an example, the guide pattern **410** may be extended along the side surface **200S** of the first semiconductor chip **200** and may have a polygonal shape, when viewed in a plan view. As another example, the guide pattern **410** may have various planar shapes described with reference to FIGS. 3A to 3D. When viewed in a plan view, the lower solder balls **510** may be provided between the first semiconductor chip **200** and the guide pattern **410**. In some example embodiments, the bumps **420** and the guide pattern **410** may be formed by an inkjet printing process or a screen printing process. The bumps **420** and the guide pattern **410** may be printed, in desired shapes, on the upper surface **200U** of the first semiconductor chip **200** and the upper surface **300U** of the first mold layer **300**.

(43) Referring to FIGS. 4 and 6, the guide pattern **410** may have the inner side surfaces **410IS**, which are disposed to face each other with the bumps **420** interposed therebetween, and outer side surfaces **410OS**, each of which is opposite to a corresponding one of the inner side surfaces **410IS**. In some example embodiments, the side surfaces **410IS** and **410OS** of the guide pattern **410** may be perpendicular to the upper surface **300U** of the first mold layer **300**, as shown in FIG. 4. In certain example embodiments, the side surfaces **410IS** and **410OS** of the guide pattern **410** may be inclined at an angle relative to the upper surface **300U** of the first mold layer **300**, as shown in FIG. 6. For example, the guide pattern **410** may be formed in such a way that its upper width is smaller than its lower width.

(44) Referring back to FIGS. 1 and 4, the guide pattern **410** may have a vertical length **410H** larger than the vertical length **420H** of each of the bumps **420**. The upper surface **410U** of the guide pattern **410** may be positioned at a height higher than a height of the upper surfaces **420U** of the bumps **420**. The inner side surfaces **410IS** of the guide pattern **410** may define a substrate mounting region **450** on the first semiconductor chip **200** and the first mold layer **300**.

(45) Referring to FIGS. 1 and 5, the second substrate **600** may be provided on the first substrate **100**. In some example embodiments, the second substrate **600** may be an interposer substrate. The first pads **610** and the second pads **620** may be provided on the upper surface **600U** and the lower surface **600L**, respectively, of the second substrate **600**. The second substrate **600** may be provided in such a way that the lower surface **600L** of the second substrate **600** faces the first semiconductor chip **200**. Upper solder balls **520** may be provided on the lower surface **600L** of the second substrate **600** and may be connected to the second pads **620**. The upper solder balls **520** may be formed of or include at least one of conductive materials (e.g., tin, lead, silver, or alloys thereof). The number or arrangement of the upper solder balls **520** may be different from the number or arrangement of the first pads **610**. As an example, the upper solder balls **520** may not be overlapped with the first pads **610**, when viewed in a plan view. The second substrate **600** may be provided on the first substrate **100** in such a way that the upper solder balls **520** are aligned to the lower solder balls **510**, respectively.

(46) The second substrate **600** may be downwardly moved to be placed in the substrate mounting region **450**. In this case, pressure **P** may be exerted on the second substrate **600** to allow the upper solder balls **520** to be in close contact with the lower solder balls **510**. For example, a dumbbell may be placed on the second substrate **600** to exert the pressure **P** on the second substrate **600**. In the case where, for close contact between the upper solder balls **520** and the lower solder balls **510**, a relatively large pressure **P** is exerted on the second substrate **600**, an electric short circuit may be formed between the upper solder balls **520**. According to some example embodiments, the bumps **420** may serve as a motion stop layer to a downward motion of the second substrate **600**. The bumps **420** may support the second substrate **600** to allow the second substrate **600** to be uniformly spaced apart from the first semiconductor chip **200** by a specific distance. Accordingly, even when a relatively large pressure is exerted on the second substrate **600**, the second substrate **600** may be supported by the bumps **420** and thus, the electric short circuit between the upper solder balls **520** may be prevented. In addition, since the second substrate **600** is placed in the substrate mounting

region **450** defined by the guide pattern **410**, misalignment between the first substrate **100** and the second substrate **600** may be prevented. As a result, the upper solder balls **520** may be easily aligned and connected to the lower solder balls **510**.

(47) Referring back to FIGS. **1** and **2**, a reflow process may be performed on the first substrate **100** and the second substrate **600**. The reflow process may be performed at a temperature higher than a melting point of the lower and upper solder balls **510** and **520**. In this case, the lower and upper solder balls **510** and **520** may be reflowed to form the solder structures **500**. The solder structures **500** may be connected to the first substrate pads **110** of the first substrate **100** and the second pads **620** of the second substrate **600**. The reflow of the lower and upper solder balls **510** and **520** may lead to an additional downward motion of the second substrate **600**. In this case, the guide pattern **410** may prevent the second substrate **600** from being laterally shifted from the first substrate **100**. Thus, the solder structures **500** may be easily formed, and the electric short circuit between the solder structures **500** may be prevented.

(48) The under-fill layer **550** may be formed between the second substrate **600** and the first semiconductor chip **200** and between the second substrate **600** and the first mold layer **300**. The under-fill layer **550** may be extended into the openings **310** in the first mold layer **300** to cover the solder structures **500**. According to some example embodiments, a thickness of a gap between the second substrate **600** and the first semiconductor chip **200** may be constantly maintained by the bumps **420**. Thus, the gap regions between the second substrate **600** and the first semiconductor chip **200** and between the second substrate **600** and the first mold layer **300** may be easily filled with the under-fill layer **550**. The first semiconductor package **P1** may be fabricated by the afore-described method, but embodiments are not limited thereto. For example, in certain example embodiments, the formation of the under-fill layer **550** may be omitted.

(49) FIGS. **7** and **8** are sectional views illustrating a method of fabricating the first semiconductor package **P1** according to an example embodiment, taken along a line corresponding to the line I-I' of FIG. **1**. For concise description, the description that follows will mainly refer to features in the fabrication method, which is different from that described with reference to FIGS. **1**, **2**, **4**, **5**, and **6**.

(50) Referring to FIGS. **7** and **8**, according to the present example embodiments, the guide pattern **410** may be formed on the upper surface **300U** of the first mold layer **300**, and the bumps **420** may be formed on the lower surface **600L** of the second substrate **600**. The bumps **420** may be formed in a region that is horizontally spaced apart from the upper solder balls **520**. The second substrate **600** may be provided in such a way that the lower surface **600L** of the second substrate **600** faces the first semiconductor chip **200**. The bumps **420** may be provided on the lower surface **600L** of the second substrate **600** to face the upper surface **200U** of the first semiconductor chip **200**. The second substrate **600** may be downwardly moved toward the substrate mounting region **450**, and in this case, the bumps **420** may be in contact with the upper surface **200U** of the first semiconductor chip **200**, as shown in FIG. **2**. The bumps **420** may support the second substrate **600** to allow the second substrate **600** to be uniformly spaced apart from the first semiconductor chip **200** by a specific distance.

(51) FIG. **9** is a sectional view illustrating a stack-type semiconductor package according to an example embodiment.

(52) Referring to FIG. **9**, a second semiconductor package **P2** may be stacked on the first semiconductor package **P1**. The first semiconductor package **P1** may be substantially the same as the first semiconductor package **P1** described with reference to FIGS. **1**, **2** and **3A** to **3D**. The second semiconductor package **P2** may include a third substrate **700**, a second semiconductor chip **800**, and a second mold layer **900**. The third substrate **700** may include upper substrate pads **710** and lower substrate pads **720**. The upper substrate pads **710** and the lower substrate pads **720** may be provided on an upper surface and a lower surface, respectively, of the third substrate **700**. The upper substrate pads **710** and the lower substrate pads **720** may be formed of or include a conductive material. Each of the lower substrate pads **720** may be connected to integrated circuits

in the third substrate **700** or a corresponding one of the upper substrate pads **710**. In FIG. **9**, a dotted line is used to depict an internal interconnection line in the third substrate **700**.

(53) As an example, the second semiconductor chip **800** may be electrically connected to the third substrate **700** through a bonding wire **820**. As another example, the second semiconductor chip **800** may be mounted on the third substrate **700** in a flip-chip manner. The second semiconductor chip **800** may be configured to perform functions different from those of the first semiconductor chip **200**. As an example, the second semiconductor chip **800** may be a memory chip. The second semiconductor chip **800** may be provided in singular or plural, and the integrated circuits in the third substrate **700** may be provided in consideration of a structure or disposition (e.g., mounting structure, kind, size, and/or number) of the second semiconductor chip **800**. Connection terminals **750** may be disposed between the second substrate **600** and the third substrate **700** and may be connected to the first pads **610** of the second substrate **600** and the lower substrate pads **720** of the third substrate **700**. The second substrate **600** may be an interposer substrate, and in this case, owing to the second substrate **600**, the disposition and number of the connection terminals **750** may be independent of the disposition or number of the solder structures **500**. The second mold layer **900** may be provided on the third substrate **700** to seal the second semiconductor chip **800**.

(54) FIG. **10** is a sectional view illustrating the first semiconductor package **P1** according to an example embodiment, taken along a line corresponding to the line I-I' of FIG. **1**. For concise description, the description that follows will mainly refer to features of the first semiconductor package **P1** different from that described with reference to FIGS. **1**, **2** and **3A** to **3D**.

(55) Referring to FIGS. **1** and **10**, a heat transfer layer **560** may be interposed between the first semiconductor chip **200** and the second substrate **600**. The heat transfer layer **560** may cover the upper surface **200U** of the first semiconductor chip **200** and may intervene between the bumps **420**. The heat transfer layer **560** may be horizontally spaced apart from the solder structures **500**. The heat transfer layer **560** may be locally provided on the upper surface **200U** of the first semiconductor chip **200**. The heat transfer layer **560** may be formed of or include a thermal interface material (TIM). Heat generated from the first semiconductor chip **200** may be easily exhausted to the outside via the heat transfer layer **560**. The bumps **420** may be spaced apart from the under-fill layer **550** by the heat transfer layer **560**. In some example embodiments, the under-fill layer **550** may be omitted.

(56) FIG. **11** is a sectional view illustrating the first semiconductor package **P1** according to an example embodiment, taken along a line corresponding to the line I-I' of FIG. **1**. For concise description, the description that follows will mainly refer to features of the first semiconductor package **P1** different from that described with reference to FIGS. **1**, **2** and **3A** to **3D**.

(57) Referring to FIGS. **1** and **11**, the guide pattern **410** may include a vertical portion **VP**, which is vertically extended from the upper surface **300U** of the first mold layer **300**, and a horizontal portion **HP**, which is extended from the vertical portion to a region between the first mold layer **300** and the second substrate **600**. The horizontal portion **HP** may be interposed between the upper surface **300U** of the first mold layer **300** and the lower surface **600L** of the second substrate **600**. The horizontal portion **HP** may be in contact with the upper surface **300U** of the first mold layer **300** and the lower surface **600L** of the second substrate **600**. The horizontal portion **HP** may be positioned at substantially the same height as the bumps **420**. The vertical portion **VP** may have a length **410H** in the direction perpendicular to the upper surface **100U** of the first substrate **100**. The length **410H** of the vertical portion **VP** may be greater than the length **420H** of each of the bumps **420**. The upper surface **410U** of the vertical portion **VP** may be positioned at a height higher than a height the upper surfaces **420U** of the bumps **420**.

(58) The vertical portion **VP** may have the inner side surfaces **410IS** facing each other with the bumps **420** interposed therebetween. The second substrate **600** may be placed between the inner side surfaces **410IS** of the vertical portion **VP**. When viewed in a plan view, the vertical portion **VP** may enclose a side surface of the second substrate **600**. The upper surface **410U** of the vertical

portion VP may be located at a height higher than the lower surface **600L** of the second substrate **600**. When viewed in a plan view, the horizontal portion HP may be extended in an extension direction of the side surface of the second substrate **600**. According to the present example embodiments, the horizontal portion HP of the guide pattern **410**, along with the bumps **420**, may serve as the motion stop layer to the downward motion of the second substrate **600** described with reference to FIG. 5. The horizontal portion HP of the guide pattern **410** may support the second substrate **600** to allow the second substrate **600** to be uniformly spaced apart from the first semiconductor chip **200** by a specific distance.

(59) FIG. 12 is a plan view illustrating the first semiconductor package P1 according to an example embodiment. FIG. 13 is a sectional view taken along a line I-I' of FIG. 12. For concise description, the description that follows will mainly refer to features of the first semiconductor package P1 different from that described with reference to FIGS. 1, 2 and 3A to 3D.

(60) Referring to FIGS. 12 and 13, the first semiconductor chip **200** may be provided on the first substrate **100**. According to the present example embodiments, the first mold layer **300** may be omitted. The solder structures **500** may be provided on the first substrate **100** and may be horizontally spaced apart from the first semiconductor chip **200**. The bumps **420** may be provided on the upper surface **200U** of the first semiconductor chip **200**. The guide pattern **410** may be disposed on the upper surface **100U** of the first substrate **100** and may be horizontally spaced apart from the first semiconductor chip **200**. The guide pattern **410** may be in contact with the upper surface **100U** of the first substrate **100**. When viewed in a plan view, the guide pattern **410** may be provided to enclose the side surface **200S** of the first semiconductor chip **200** and the solder structures **500** may be provided between the first semiconductor chip **200** and the guide pattern **410**.

(61) The guide pattern **410** may protrude from the upper surface **100U** of the first substrate **100**. The guide pattern **410** may have a length **410H** in the direction perpendicular to the upper surface **100U** of the first substrate **100**. The length **410H** of the guide pattern **410** may be greater than the length **420H** of each of the bumps **420**. The upper surface **410U** of the guide pattern **410** may be positioned at a height higher than a height of the upper surfaces **420U** of the bumps **420**. A lower surface **410L** of the guide pattern **410** may be located at substantially the same height as the upper surface **100U** of the first substrate **100**. The guide pattern **410** may have inner side surfaces **410IS**, which are provided to face each other with the bumps **420** interposed therebetween. The second substrate **600** may be provided on the first semiconductor chip **200**. The second substrate **600** may be disposed between the inner side surfaces **410IS** of the guide pattern **410**. When viewed in a plan view, the guide pattern **410** may be provided to enclose the side surface of the second substrate **600**. The upper surface **410U** of the guide pattern **410** may be located at a height higher than the lower surface **600L** of the second substrate **600**.

(62) The heat transfer layer **560** may be interposed between the first semiconductor chip **200** and the second substrate **600**. The heat transfer layer **560** may cover the upper surface **200U** of the first semiconductor chip **200** and may intervene between the bumps **420**. The heat transfer layer **560** may be substantially the same as the heat transfer layer **560** described with reference to FIG. 10. In some example embodiments, the under-fill layer **550** may be provided to fill a gap region between the first substrate **100** and the second substrate **600**. In certain example embodiments, the under-fill layer **550** may be omitted.

(63) FIG. 14 is a sectional view illustrating a stack-type semiconductor package according to an example embodiment.

(64) Referring to FIG. 14, the second semiconductor package P2 may be stacked on the first semiconductor package P1. Except for the absence of the second substrate **600**, the first semiconductor package P1 may be substantially the same as the first semiconductor package P1 described with reference to FIGS. 1, 2 and 3A to 3D. The second semiconductor package P2 may include the third substrate **700**, the second semiconductor chip **800**, and the second mold layer **900**.

The third substrate **700** may include the upper substrate pads **710** and the lower substrate pads **720**. The second semiconductor package **P2** may be substantially the same as the second semiconductor package **P2** described with reference to FIG. **9**. According to the present example embodiments, the third substrate **700** may be provided on the first semiconductor chip **200** and the first mold layer **300**. The solder structures **500** may connect the first substrate pads **110** of the first substrate **100** to the lower substrate pads **720** of the third substrate **700**. The third substrate **700** may be electrically connected to the first substrate **100** and the first semiconductor chip **200** through the solder structures **500**. The bumps **420** may be interposed between the first semiconductor chip **200** and the third substrate **700**. The bumps **420** may be in contact with the upper surface **200U** of the first semiconductor chip **200** and a lower surface **700L** of the third substrate **700**. The inner side surfaces **410IS** of the guide pattern **410** may be provided to face each other with the bumps **420** interposed therebetween. The third substrate **700** may be provided between the inner side surfaces **410IS** of the guide pattern **410**. When viewed in a plan view, the guide pattern **410** may be provided to enclose a side surface of the third substrate **700**. The upper surface **410U** of the guide pattern **410** may be located at a height higher than the lower surface **700L** of the third substrate **700**.

(65) In some example embodiments, the under-fill layer **550** may be interposed between the third substrate **700** and the first semiconductor chip **200** and between the third substrate **700** and the first mold layer **300**. In certain example embodiments, the under-fill layer **550** may be omitted.

According to the present example embodiments, the bumps **420** may support the third substrate **700** to allow the third substrate **700** to be uniformly spaced apart from the first semiconductor chip **200** by a specific distance. In addition, since the third substrate **700** is placed in the substrate mounting region defined by the inner side surfaces **410IS** of the guide pattern **410**, the first semiconductor package **P1** and the second semiconductor package **P2** may be prevented from being misaligned to each other. Thus, the solder structures **500** may be easily formed, and the electric short circuit between the solder structures **500** may be prevented.

(66) FIG. **15** is a plan view illustrating the first semiconductor package **P1** according to an example embodiment. FIG. **16** is a sectional view taken along a line I-I' of FIG. **15**, and FIG. **17** is an enlarged view illustrating a portion 'AA' of FIG. **16**.

(67) Referring to FIGS. **15** to **17**, the first semiconductor chip **200** may be mounted on the first substrate **100**. The first substrate **100** may be a printed circuit board with a circuit pattern or a redistribution layer. The first substrate **100** may include the first substrate pads **110** and the second substrate pads **120**. The first substrate pads **110** may be provided on the upper surface **100U** of the first substrate **100**, and the second substrate pads **120** may be provided on the lower surface **100L** of the first substrate **100**. The second substrate pads **120** may be connected to corresponding pads of the first substrate pads **110** through internal interconnection lines in the first substrate **100**. The first substrate pads **110** and the second substrate pads **120** may be formed of or include at least one of conductive materials. The outer terminals **130** may be provided on the lower surface **100L** of the first substrate **100** and may be connected to the second substrate pads **120**. The outer terminals **130** may be formed of or include at least one of conductive materials and may be provided in the form of solder balls.

(68) The first semiconductor chip **200** may include the chip pads **210** provided on a lower surface thereof. The connecting portions **220** may be provided between the first substrate **100** and the first semiconductor chip **200** and may be connected to the chip pads **210**, respectively. The connecting portions **220** may be connected to corresponding pads of the first substrate pads **110**. The first semiconductor chip **200** may be electrically connected to the first substrate **100** through the connecting portions **220**. The chip pads **210** and the connecting portions **220** may be formed of or include at least one of conductive materials. Each of the connecting portions **220** may include one of solder balls, bumps, or pillars. The first semiconductor chip **200** may include integrated circuits, and the integrated circuits may be adjacent to the lower surface of the first semiconductor chip **200**.

The first semiconductor chip **200** may be a logic chip. As an example, the first semiconductor chip **200** may be a non-memory chip (e.g., an application processor).

(69) The second substrate **600** may be provided on the first substrate **100**, and the first semiconductor chip **200** may be provided between the first substrate **100** and the second substrate **600**. As an example, the second substrate **600** may be an interposer substrate and may be formed of or include at least one of insulating resins. The insulating resin may include a solder resist material (e.g., photosensitive polyimide), but embodiments are not limited to this example. The first pads **610** may be provided on the upper surface **600U** of the second substrate **600**, and the second pads **620** may be provided on the lower surface **600L** of the second substrate **600**. Interconnection lines in the second substrate **600** may connect the first pads **610** and the second pads **620** electrically to each other. The first pads **610** and the second pads **620** may be formed of or include at least one of conductive or metallic materials (e.g., copper or aluminum).

(70) The solder structures **500** may be provided between the first substrate **100** and the second substrate **600** and may be horizontally spaced apart from the first semiconductor chip **200**. The solder structures **500** may be connected to corresponding pads of the first substrate pads **110**. Each of the solder structures **500** may be electrically connected to the outer terminal **130** or the first semiconductor chip **200**. The solder structures **500** may be connected to the second pads **620**. The second substrate **600** may be electrically connected to the first substrate **100** and the first semiconductor chip **200** through the solder structures **500**. The solder structures **500** may be formed of or include at least one of conductive materials (e.g., tin, lead, silver, or alloys thereof).

(71) A plurality of supporting patterns **460** may be provided between the first substrate **100** and the second substrate **600** and may be horizontally spaced apart from the first semiconductor chip **200**. The solder structures **500** may be provided between the first semiconductor chip **200** and the supporting patterns **460**. When viewed in a plan view, the supporting patterns **460** may be arranged to enclose the first semiconductor chip **200**. For example, when viewed in a plan view, the supporting patterns **460** may be arranged along the side surfaces **200S** of the first semiconductor chip **200**. At least two patterns of the supporting patterns **460** may be spaced apart from each other in an extension direction of one of the side surfaces **200S** of the first semiconductor chip **200**, when viewed in a plan view. The supporting patterns **460** may have a pillar shape and protrude from the upper surface **100U** of the first substrate **100**.

(72) An outer side surface **460SO** of each of the supporting patterns **460** may be aligned to a side surface **100S** of the first substrate **100** and a side surface **600S** of the second substrate **600**. Each of the supporting patterns **460** may have an inner side surface **460SI** facing the outer side surface **460SO**. In some example embodiments, the inner side surface **460SI** of each of the supporting patterns **460** may be inclined at an angle relative to the upper surface **100U** of the first substrate **100**, when viewed in a sectional view. In this case, each of the supporting patterns **460** may be formed in such a way that its upper width is smaller than its lower width. In some example embodiments, at least one of the supporting patterns **460** may be spaced apart from the lower surface **600L** of the second substrate **600**. For example, each of the supporting patterns **460** may be spaced apart from the lower surface **600L** of the second substrate **600**. The supporting patterns **460** may be formed of or include at least one of insulating materials (e.g., insulating polymer materials or solder resist materials).

(73) The bumps **420** may be provided between the first semiconductor chip **200** and the second substrate **600** and may be horizontally spaced apart from each other on the upper surface **200U** of the first semiconductor chip **200**. The bumps **420** may have a pillar shape and extend from the upper surface **200U** of the first semiconductor chip **200** to the lower surface **600L** of the second substrate **600**. The bumps **420** may be in contact with the upper surface **200U** of the first semiconductor chip **200** and the lower surface **600L** of the second substrate **600**. In some example embodiments, the bumps **420** may be formed of or include the same material as the second substrate **600**. In certain example embodiments, the bumps **420** may be formed of or include the

same material as the supporting patterns **460**. The bumps **420** may include at least one of insulating polymer materials, conductive polymer materials, or solder resist materials.

(74) The first mold layer **300** may be provided between the first substrate **100** and the second substrate **600** to cover the first semiconductor chip **200** and the solder structures **500**. The first mold layer **300** may be provided to fill a gap region between the first substrate **100** and the first semiconductor chip **200** and to seal the connecting portions **220**. The first mold layer **300** may be provided to fill a gap region between the second substrate **600** and the first semiconductor chip **200** and may be interposed between the bumps **420**. The first mold layer **300** may be provided to cover the inner side surface **460SI** of each of the supporting patterns **460**. The first mold layer **300** may be extended between at least one of the supporting patterns **460** and the lower surface **600L** of the second substrate **600** and may be interposed therebetween. As an example, the first mold layer **300** may be interposed between the lower surface **600L** of the second substrate **600** and each of the supporting patterns **460**. The first mold layer **300** may be formed of or include at least one of epoxy molding compounds or insulating resins.

(75) FIG. **18** is a sectional view illustrating a modified structure of the first semiconductor package **P1** according to an example embodiment, taken along a line I-I' of FIG. **15**. FIG. **19** is an enlarged view of a portion a portion 'BB' of FIG. **18**. For concise description, the description that follows will mainly refer to features of the first semiconductor package **P1** different from that described with reference to FIGS. **15** to **17**.

(76) Referring to FIGS. **15**, **18**, and **19**, the supporting patterns **460** according to the present example embodiments may be in contact with the lower surface **600L** of the second substrate **600**. Thus, the first mold layer **300** may not be interposed between the supporting patterns **460** and the lower surface **600L** of the second substrate **600**.

(77) FIG. **20** is a plan view illustrating a method of fabricating the first semiconductor package **P1** according to an example embodiment. FIGS. **21** to **24** are sectional views illustrating a method of fabricating the first semiconductor package **P1** according to an example embodiment, taken along a line corresponding to a line I-I' of FIG. **20**. FIG. **25** is an enlarged view of a portion CC of FIG. **21**, FIG. **26** is an enlarged view of a portion DD of FIG. **23**, and FIG. **27** is an enlarged view of a portion EE of FIG. **24**. In order to avoid redundancy, overlapping description of the first semiconductor package **P1** described with reference to FIGS. **15** to **19** may be omitted.

(78) Referring to FIGS. **20**, **21**, and **25**, the first substrate **100** including a mount region **R1** and a dummy region **R2** may be provided. The mount region **R1** may be a region of the first substrate **100**, on which the first semiconductor chip **200** is mounted, and the dummy region **R2** may be another region of the first substrate **100**, which will be removed by a subsequent sawing process. In some example embodiments, the first substrate **100** may include a plurality of mount regions **R1**, and the dummy region **R2** may be provided to enclose each of the mount regions **R1**. The dummy region **R2** may be interposed between adjacent ones of the mount regions **R1**.

(79) A plurality of first semiconductor chips **200** may be respectively mounted on the mount regions **R1** of the first substrate **100**. Each of the first semiconductor chips **200** may include the chip pads **210** provided on a lower surface thereof. The connecting portions **220** may be provided between the first substrate **100** and each of the first semiconductor chips **200** and may be connected to the chip pads **210**, respectively. Each of the first semiconductor chips **200** may be electrically connected to the first substrate **100** through the connecting portions **220**. The lower solder balls **510** may be formed on each of the mount regions **R1** of the first substrate **100**. The lower solder balls **510** may be electrically connected to the outer terminals **130** or the first semiconductor chips **200**. The lower solder balls **510** may be formed of or include at least one of conductive materials (e.g., tin, lead, silver, or alloys thereof).

(80) A plurality of the supporting patterns **460** may be formed on the first substrate **100**. The supporting patterns **460** may include first supporting patterns **460a**, which are formed on a border **RB** between the dummy region **R2** and each of the mount regions **R1**, and second supporting

patterns **460b**, which are formed on the dummy region **R2**. When viewed in a plan view, the first supporting patterns **460a** may enclose each of the first semiconductor chips **200**. The lower solder balls **510** may be provided between the first supporting patterns **460a** and each of the first semiconductor chips **200**. When viewed in a plan view, the second supporting patterns **460b** may enclose each of the first semiconductor chips **200**, and the first supporting patterns **460a** may be provided between the second supporting patterns **460b** and each of the first semiconductor chips **200**. In some example embodiments, the formation of the second supporting patterns **460b** may be omitted.

(81) The supporting patterns **460** may be formed by an inkjet printing process or a screen printing process. The supporting patterns **460** may be printed in a desired pattern shape on the upper surface **100U** of the first substrate **100**. The supporting patterns **460** may have a pillar shape and protrude from the upper surface **100U** of the first substrate **100**. In some example embodiments, each of the supporting patterns **460** may be formed in such a way that its upper width is smaller than its lower width. The supporting patterns **460** may be formed of or include at least one of insulating materials (e.g., insulating polymer materials or solder resist materials).

(82) Referring to FIGS. **20** and **22**, a plurality of second substrates **600** may be provided on the first substrate **100**. The second substrates **600** may be used as interposer substrates. The second substrates **600** may be overlapped with the mount regions **R1**, respectively, of the first substrate **100**. The first pads **610** may be provided on the upper surface **600U** of each of the second substrates **600**, and the second pads **620** may be provided on the lower surface **600L** of each of the second substrates **600**. Each of the second substrates **600** may be provided in such a way that the lower surface **600L** thereof faces a corresponding one of the first semiconductor chip **200**. The upper solder balls **520** may be provided on the lower surface **600L** of each of the second substrates **600** and may be connected to the second pads **620**. The upper solder balls **520** may be formed of or include at least one of conductive materials (e.g., tin, lead, silver, or alloys thereof). Each of the second substrates **600** may be provided on the first substrate **100** in such a way that the upper solder balls **520** are aligned to the lower solder balls **510**, respectively.

(83) A plurality of the bumps **420** may be formed on the lower surface **600L** of each of the second substrates **600**. The bumps **420** may be formed in a region that is horizontally spaced apart from the upper solder balls **520**. The bumps **420** may be provided to face the upper surface **200U** of each of the first semiconductor chips **200**. The bumps **420** may have a pillar shape and protrude from the lower surface **600L** of each of the second substrates **600**. In some example embodiments, the bumps **420** may be formed of or include the same material as each of the second substrates **600**. In certain example embodiments, the bumps **420** may be formed of the same material as the supporting patterns **460**. In this case, the bumps **420** may be formed by an inkjet printing process or a screen printing process.

(84) Referring to FIGS. **20**, **23**, and **26**, the second substrates **600** may be downwardly moved onto the mount regions **R1**, respectively, of the first substrate **100**. In this case, pressure **P** may be exerted on the second substrates **600** to allow the upper solder balls **520** to be in close contact with the lower solder balls **510**. For example, a dumbbell **650** may be provided on each of the second substrates **600** to exert the pressure **P** on each of the second substrates **600**. During the downward motion of each of the second substrates **600**, the bumps **420** and the supporting patterns **460** may support the second substrates **600**. According to the present example embodiments, the supporting patterns **460** may support the second substrates **600** to allow each of the second substrates **600** to be uniformly spaced apart from the first substrate **100** by a specific distance. Thus, even when a relatively large pressure **P** is exerted on each of the second substrates **600**, it may be possible to prevent an edge region of each of the second substrates **600** from being bent toward the first substrate **100**.

(85) Heat **H** may be applied to the first substrate **100** and the second substrates **600** to allow the upper solder balls **520** and the lower solder balls **510** to form junctions. The heat **H** may be applied

to the first substrate **100** and the second substrates **600**, when the pressure **P** is exerted on the second substrates **600**. As an example, the second substrates **600** may be bonded to the first substrate **100** by a thermal compression bonding method. As the upper solder balls **520** and the lower solder balls **510** form the junctions, the solder structures **500** may be formed. The solder structures **500** may be connected to the first substrate pads **110** of the first substrate **100** and the second pads **620** of each of the second substrates **600**.

(86) After the process of bonding each of the second substrates **600** on the first substrate **100**, the dumbbell **650** may be removed. Depending on a vertical height **460H** of the supporting patterns **460**, the supporting patterns **460** may be separated from or in contact with the lower surface **600L** of each of the second substrates **600**, after the removal of the dumbbell **650**. For example, in the case where the vertical height **460H** of the supporting patterns **460** is relatively small, the supporting patterns **460** may be spaced apart from the lower surface **600L** of each of the second substrates **600**, after the removal of the dumbbell **650**. By contrast, in the case where the vertical height **460H** of the supporting patterns **460** is relatively large, the supporting patterns **460** may be in contact with the lower surface **600L** of each of the second substrates **600**, after the removal of the dumbbell **650**, as shown in FIG. 26.

(87) Referring to FIGS. 20, 24, and 27, the first mold layer **300** may be formed to fill a gap region between the first substrate **100** and the second substrates **600** and a gap region between the second substrates **600**. The first mold layer **300** may expose the upper surfaces **600U** of the second substrates **600**. In some example embodiments, the supporting patterns **460** may be spaced apart from the lower surface **600L** of each of the second substrates **600**. In this case, the first mold layer **300** may cover the side surfaces of the supporting patterns **460** and may be extended into regions between the supporting patterns **460** and the lower surface **600L** of each of the second substrates **600**. In certain example embodiments, as shown in FIG. 27, the supporting patterns **460** may be in contact with the lower surface **600L** of each of the second substrates **600**, and the first mold layer **300** may cover the side surfaces of the supporting patterns **460**.

(88) A stack **SS** including a plurality of the first semiconductor packages **P1** may be formed, as a result of the afore-described processes. A sawing process **SP** may be performed on the stack **SS** to divide the stack **SS** into the plurality of the first semiconductor packages **P1**. The dummy region **R2** of the first substrate **100** and structures on the dummy region **R2** may be removed by the sawing process **SP**. As an example, the second supporting patterns **460b** of the supporting patterns **460** may be removed by the sawing process **SP**. The first supporting patterns **460a** of the supporting patterns **460** may be formed on the border **RB** between the dummy region **R2** and each of the mount regions **R1** of the first substrate **100**. Thus, a portion of each of the first supporting patterns **460a**, which is located outside the border **RB**, may be removed by the sawing process **SP**. After the sawing process **SP**, a portion of each of the first supporting patterns **460a**, which is located inside the border **RB**, may remain on a corresponding one of the mount region **R1** and may constitute a corresponding one of the first semiconductor package **P1**.

(89) FIG. 28 is a sectional view illustrating a method of fabricating the first semiconductor package **P1** according to a modified example embodiment, taken along a line corresponding to the line I-I' of FIG. 20. For concise description, the description that follows will mainly refer to features in the fabrication method, which is different from that described with reference to FIGS. 20 to 27.

(90) Referring to FIG. 28, the bumps **420** may be formed on the upper surface **200U** of each of the first semiconductor chips **200**. The bumps **420** may have a pillar shape and protrude from the upper surface **200U** of each of the first semiconductor chips **200**. According to the present example embodiments, the bumps **420** may be formed of or include the same material as the supporting patterns **460**. The bumps **420** may be formed by an inkjet printing process or a screen printing process.

(91) FIG. 29 is a plan view illustrating a method of fabricating a first semiconductor package according to an example embodiment. FIGS. 30 to 33 are sectional views illustrating a method of

fabricating the first semiconductor package **P1** according to an example embodiment, taken along a line corresponding to a line I-I' of FIG. **29**. For concise description, the description that follows will mainly refer to features in the fabrication method, which is different from that described with reference to FIGS. **20** to **27**.

(92) Referring to FIGS. **29** and **30**, the first substrate **100** including the mount region **R1** and the dummy region **R2** may be provided. According to the present example embodiments, the first substrate **100** may include a plurality of the mount regions **R1**, which are provided adjacent to each other, and the dummy region **R2** may be provided to enclose the plurality of the mount regions **R1**. Each of the mount regions **R1** may be directly connected to others of the mount regions **R1** adjacent thereto. In other words, the dummy region **R2** may not be interposed between adjacent ones of mount regions **R1**. The first semiconductor chips **200** may be mounted on the mount regions **R1**, respectively, of the first substrate **100**, and the lower solder balls **510** may be formed on each of the mount regions **R1** of the first substrate **100**.

(93) The supporting patterns **460** may be formed on the first substrate **100**. The supporting patterns **460** may include the first supporting patterns **460a** and the second supporting patterns **460b**. According to the present example embodiments, the first supporting patterns **460a** may be formed on the border **RB** between the dummy region **R2** and each of the mount regions **R1** and between adjacent ones of mount regions **R1**, and the second supporting patterns **460b** may be formed on the dummy region **R2**. When viewed in a plan view, the first supporting patterns **460a** may enclose each of the first semiconductor chips **200**. At least some of the first supporting patterns **460a** may be formed between adjacent ones of the first semiconductor chips **200**. The lower solder balls **510** may be provided between the first supporting patterns **460a** and each of the first semiconductor chips **200**. When viewed in a plan view, the second supporting patterns **460b** may enclose the first semiconductor chips **200**, and the first supporting patterns **460a** may be formed between the second supporting patterns **460b** and each of the first semiconductor chips **200**. In some example embodiments, the formation of the second supporting patterns **460b** may be omitted.

(94) Referring to FIGS. **29** and **31**, according to the present example embodiments, one second substrate **600** may be provided on the first substrate **100**. The second substrate **600** may be an interposer substrate. The second substrate **600** may be overlapped with the mount regions **R1** of the first substrate **100**. The upper solder balls **520** may be provided on the lower surface **600L** of the second substrate **600**. The second substrate **600** may be provided on the first substrate **100** in such a way that the upper solder balls **520** are aligned to the lower solder balls **510**, respectively. The bumps **420** may be provided on the lower surface **600L** of the second substrate **600**. The bumps **420** may be provided to face the upper surface **200U** of each of the first semiconductor chips **200**.

(95) Referring to FIGS. **29** and **32**, the second substrate **600** may be downwardly moved toward the mount regions **R1** of the first substrate **100**. The pressure **P** may be exerted on the second substrate **600** to allow the upper solder balls **520** to be in close contact with the lower solder balls **510**. During the downward motion of the second substrate **600**, the bumps **420** and the supporting patterns **460** may support the second substrates **600**. According to the present example embodiments, the supporting patterns **460** may support the second substrate **600** to allow the second substrate **600** to be uniformly spaced apart from the first substrate **100** by a specific distance. Thus, even when a relatively large pressure **P** is exerted on the second substrate **600**, it may be possible to prevent an edge region of the second substrate **600** from being bent toward the first substrate **100**. As the upper solder balls **520** and the lower solder balls **510** form junctions, the solder structures **500** may be formed.

(96) Referring to FIGS. **29** and **33**, the first mold layer **300** may fill a gap region between the first substrate **100** and the second substrate **600**.

(97) As a result of the afore-described processes, the stack **SS** including a plurality of the first semiconductor packages **P1** may be formed. A sawing process **SP** may be performed on the stack **SS**, and as a result, the stack **SS** may be divided into the plurality of the first semiconductor

packages **P1**. The dummy region **R2** of the first substrate **100** and structures on the dummy region **R2** may be removed by the sawing process **SP**. As an example, the second supporting patterns **460b** of the supporting patterns **460** may be removed by the sawing process **SP**. According to the present example embodiments, each of the first supporting patterns **460a** of the supporting patterns **460** may be cut by the sawing process **SP**. As an example, the first supporting patterns **460a** may include a first pattern, which is located on the border **RB** between the dummy region **R2** and the mount region **R1** and thus is at least partially removed by the sawing process **SP**. After the sawing process **SP**, at least a portion of the first pattern of the first supporting patterns **460a** may remain on a corresponding one of the mount region **R1** and may constitute a corresponding one of the first semiconductor package **P1**. In addition, the first supporting patterns **460a** may include a second pattern, which is located on the border **RB** between the mount regions **R1** and is cut by the sawing process **SP**. In this case, at least a portion of the second pattern of the first supporting patterns **460a** may remain on a corresponding one of the mount regions **R1** and may constitute a corresponding one of the first semiconductor packages **P1**.

(98) FIG. **34** is a sectional view illustrating a stack-type semiconductor package according to an example embodiment.

(99) Referring to FIG. **34**, the second semiconductor package **P2** may be stacked on the first semiconductor package **P1**. The first semiconductor package **P1** may be substantially the same as the first semiconductor package **P1** described with reference to FIGS. **15** to **19**. The second semiconductor package **P2** may include the third substrate **700**, the second semiconductor chip **800**, and the second mold layer **900**. The third substrate **700** may include the upper substrate pads **710** and the lower substrate pads **720**. The upper substrate pads **710** and the lower substrate pads **720** may be provided on an upper surface and a lower surface, respectively, of the third substrate **700**. The upper substrate pads **710** and the lower substrate pads **720** may be formed of or include a conductive material. Each of the lower substrate pads **720** may be connected to integrated circuits in the third substrate **700** or a corresponding one of the upper substrate pads **710**. In FIG. **34**, a dotted line is used to depict an internal interconnection line in the third substrate **700**. As an example, the second semiconductor chip **800** may be electrically connected to the third substrate **700** through the bonding wire **820**. As another example, the second semiconductor chip **800** may be mounted on the third substrate **700** in a flip-chip manner. The second semiconductor chip **800** may be configured to perform functions different from those of the first semiconductor chip **200**. As an example, the second semiconductor chip **800** may be a memory chip. The connection terminals **750** may be provided between the second substrate **600** and the third substrate **700** and may be connected to the first pads **610** of the second substrate **600** and the lower substrate pads **720** of the third substrate **700**. The second substrate **600** may be an interposer substrate. The second mold layer **900** may be provided on the third substrate **700** to seal the second semiconductor chip **800**.

(100) According to some example embodiments, a semiconductor package having a minimized number of defects and improved reliability and a method of fabricating the same may be provided.

(101) While example embodiments have been particularly shown and described, it will be understood by one of ordinary skill in the art that variations in form and detail may be made therein without departing from the spirit and scope of the attached claims.

Claims

1. A semiconductor package comprising: a first substrate; a second substrate provided on the first substrate; a first semiconductor chip provided between the first substrate and the second substrate; solder structures provided between the first substrate and the second substrate and spaced apart from the first semiconductor chip, the solder structures electrically connecting the first substrate and the second substrate; and supporting patterns provided between the first substrate and the second substrate, wherein the solder structures are disposed between the first semiconductor chip

and the supporting patterns, wherein, in a plan view of the semiconductor package, the supporting patterns are spaced apart from each other around sides of the first semiconductor chip, and wherein the solder structures comprise a conductive material, and the supporting patterns comprise an insulating material.

2. The semiconductor package of claim 1, wherein the supporting patterns are spaced apart from the first semiconductor chip and the solder structures.

3. The semiconductor package of claim 1, wherein the supporting patterns are arranged to surround the sides of the first semiconductor chip.

4. The semiconductor package of claim 1, further comprising a first mold layer provided between the first substrate and the second substrate and covering the first semiconductor chip, wherein the first mold layer covers inner side surfaces of the supporting patterns.

5. The semiconductor package of claim 4, wherein the first mold layer extends between at least one of the supporting patterns and a lower surface of the second substrate.

6. The semiconductor package of claim 4, wherein at least one of the supporting patterns is in contact with a lower surface of the second substrate.

7. The semiconductor package of claim 4, wherein the inner side surfaces of the supporting patterns are inclined at an angle relative to an upper surface of the first substrate.

8. The semiconductor package of claim 4, wherein the supporting patterns has outer side surfaces opposite to the inner side surfaces, and wherein the outer side surfaces of the supporting patterns are aligned with side surfaces of the first substrate.

9. The semiconductor package of claim 8, wherein the outer side surfaces of the supporting patterns are aligned with side surfaces of the second substrate.

10. The semiconductor package of claim 1, further comprising bumps disposed between the first semiconductor chip and the second substrate, wherein the bumps are spaced apart from each other, between the first semiconductor chip and the second substrate.

11. The semiconductor package of claim 10, wherein the bumps and the second substrate comprise a same material.

12. The semiconductor package of claim 10, wherein the bumps and the supporting patterns comprise a same material.

13. The semiconductor package of claim 10, wherein each of the bumps has a pillar shape and protrudes from an upper surface of the first semiconductor chip toward a lower surface of the second substrate.

14. A semiconductor package comprising: a first substrate; a second substrate provided on the first substrate; a first semiconductor chip provided between the first substrate and the second substrate, the first semiconductor chip including chip pads on a lower surface thereof; connecting portions provided between the first substrate and the lower surface of the first semiconductor chip, the connecting portions connected to the chip pads; solder structures provided between the first substrate and the second substrate and spaced apart from the first semiconductor chip, the solder structures electrically connecting the first substrate and the second substrate; and supporting patterns provided between the first substrate and the second substrate, wherein the solder structures are disposed between the first semiconductor chip and the supporting patterns, wherein, in a plan view of the semiconductor package, the supporting patterns are spaced apart from each other around sides of the first semiconductor chip, and wherein the solder structures comprise a conductive material, and the supporting patterns comprise an insulating material.

15. The semiconductor package of claim 14, further comprising a first mold layer provided between the first substrate and the second substrate and covering the first semiconductor chip, wherein the first mold layer covers inner side surfaces of the supporting patterns.

16. The semiconductor package of claim 15, wherein the first mold layer extends between at least one of the supporting patterns and a lower surface of the second substrate.

17. The semiconductor package of claim 15, wherein at least one of the supporting patterns is in

contact with a lower surface of the second substrate.

18. The semiconductor package of claim 15, wherein the supporting patterns has outer side surfaces opposite to the inner side surfaces, and wherein the outer side surfaces of the supporting patterns are aligned with side surfaces of the first substrate.

19. The semiconductor package of claim 1, wherein each of the supporting patterns comprises a pillar shape.

20. The semiconductor package of claim 4, wherein, in the plan view, adjacent ones of the supporting patterns around the sides of the first semiconductor chip are separated from each other by the first mold layer.
